

A design that looks right on screen can still be the wrong size in real life.

proportion • ratio • ruler • scale

1. A design's size relative to its real, actual-size object or space is called _____, usually written as a _____ such as 1:2 or 1:10.
2. If a scale is 1:10, one unit on the drawing equals _____ real units. If a real box is 20 inches wide and you're drawing it at a scale of 1:4, your drawing should be _____ inches wide.
3. Keeping the size relationship between different parts of a design consistent when you scale it up or down is called _____. Scaling only the width and not the height distorts this relationship.
4. A ruler with several sets of graduated markings, one per scale ratio, used to measure or draw at scale without doing the math for every measurement, is a _____.
5. **Practice.** A product's actual packaging is 8" wide × 12" tall. You need to draw it at a scale of 1:2 for a presentation board. What are the scaled dimensions?

6. **Practice.** A poster's actual size is 24" × 36". Your sketchbook thumbnail is drawn at 4" × 6". What scale ratio is that thumbnail drawn at?

Discuss: Where does scale matter most in a designer's work — packaging, signage, print sizing, something else?